Monish Prasanna S

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Summary

Computer Science (Cyber Security) student with a strong foundation in data structures, algorithms, and object-oriented design, combined with hands-on experience building secure infrastructure, cloud-native applications, and embedded systems. Comfortable working across the stack – from low-level firmware to serverless cloud architecture – with a practical, build-it-and-break-it approach to problem solving.

Education

Amrita Vishwa Vidyapeetham Aug 2023 – Present
B.Tech, Computer Science and Engineering (Cyber Security) – CGPA: 7.31 *Expected Graduation: 2027*

Technical Skills

Languages: Python, C++, C
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Computer Networks
Cloud & Systems: Cloudflare Workers, Docker, Linux Administration, Tailscale VPN, Shell Scripting,Kubernetes(K8S)
Security Tools: Burp Suite, Wireshark, Malware Analysis, Portswigger
Other: Git, REST APIs, Android/iOS App Development

Projects

URL Shortener | *Cloudflare Workers, Cloudflare KV, Edge Routing, Tailwind CSS* | Live
- Designed a hash-based key generation scheme and deployed a live, production redirect engine on Cloudflare's serverless edge network at zero infrastructure cost.
- Implemented O(1) average-case lookups against a distributed KV store, supporting custom string aliases and live click-tracking analytics via a minimalist terminal-style dashboard.

Self-Hosted Home Server Infrastructure | *Linux, Docker, Tailscale VPN, Network Virtualization*
- Converted x86 hardware into a self-hosted home server, containerizing and managing 3 isolated services (qBittorrent, n8n, and a media streaming service) via Docker with independent resource allocation.
- Configured a zero-trust Tailscale overlay network mesh with custom system domains and persistent routing for secure, low-latency remote access.

USB HID Trust Gaps in Modern Operating Systems | *Embedded C/C++, ATtiny85, Keystroke Injection*
- Engineered low-cost HID implants using ATtiny85 microcontrollers and the V-USB stack to evaluate physical I/O trust gaps against hardened Windows enterprise security baselines.
- Automated Wi-Fi credential exfiltration via a webhook-based payload, demonstrating a practical attack chain from firmware to network callback.

Web Application Vulnerability Analysis | *Bugcrowd, SonarQube, GitHub Actions, PortSwigger*
- Conducted structured vulnerability reporting on Bugcrowd with a focus on identifying and documenting CORS misconfigurations.
- Automated early vulnerability detection by integrating SonarQube static analysis (SAST) directly into GitHub Actions CI/CD workflows.
- Completed advanced web security learning labs on PortSwigger Web Security Academy to enhance practical exploitation and auditing skills.

Evasive Network Tunneling & Server Hosting | *Docker, Chisel, HTTP Tunneling, TCP/UDP*
- Deployed and containerized a multiplayer game server using Docker within a highly restricted enterprise campus network.
- Bypassed strict firewall policies blocking standard TCP/UDP traffic by utilizing Chisel to encapsulate and tunnel routing packets over allowed HTTP/WebSocket protocols.
- Engineered a reliable client-server port-forwarding architecture, restoring seamless local network connectivity and multiplayer access without triggering network alarms.

Certifications

Amazon Web Service (AWS): Certified Cloud Practitioner
Palo Alto Networks: Cybersecurity, Network, Cloud, and Security Operations Fundamentals
Cisco: Introduction to Cyber Security, Cisco Packet Tracer

Publications

Evaluating USB HID Trust Gaps in Modern Operating Systems Sept 2026
International Conference on Advancement in Cyber Security and Digital Forensics (ICACSDF 2026), UPES *Dehradun, India*
- Presented original research (Paper ID 808) on physical I/O trust gaps in hardened OS environments, co-organized by IEEE UP Section and Anusandhan National Research Foundation.

Volunteering & Leadership

CTF Contributor
NULLBORN
- Contributed challenges to the main CTF event managed by the Nullborn community.

Organizer
Morphx – Amrita CyberNation 2025
- Designed reverse-engineering and vulnerability-hunting challenges, requiring teams to deconstruct pre-built systems and trace attack chains.